

The empirical recurrence for OEIS A229636, recovered from its tail

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8 September 2026

Abstract

OEIS A229636 records “Empirical recurrence of order 57” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 412$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 61$. Its last coefficient is $c_{57} = 32768 \neq 0$, so no further tail satisfies anything shorter; any order-57 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A229636 is “Number of defective 3-colorings of an $n \times 7$ array connected horizontally, diagonally and antidiagonally with exactly two mistakes, and colors introduced in row-major 0..2 order.”. It has offset 1 and begins

120, 5392, 136778, 3723370, 98015235, 2495874984, 61836040854, 1498317588826, ...

The entry states:

Empirical recurrence of order 57 (see link above).

As of the “Last modified” line on the live entry (Apr 27 2021 22:20:27, revision 8) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 412$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 412$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 57: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 57.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 824$ exact terms from $n = 4$ on, it returns order 57 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{57} c_i a(n-i),$$

c_1	123	c_{16}	551217620746407	c_{31}	−804212022256406694	c_{46}	46012304097474
c_2	−6690	c_{17}	316551389019024	c_{32}	−187152208156347708	c_{47}	−24591298047858
c_3	211829	c_{18}	−4078611361929752	c_{33}	800178243628357779	c_{48}	2023150285956
c_4	−4314738	c_{19}	4256973077801529	c_{34}	−335441264618712678	c_{49}	914890479012
c_5	58615974	c_{20}	14061931485832206	c_{35}	−322039127587847922	c_{50}	−237485278536
c_6	−524537364	c_{21}	−34872564483562048	c_{36}	363730185484923424	c_{51}	3749561784
c_7	2783886546	c_{22}	−12980924700082719	c_{37}	−40886775854647428	c_{52}	6141890832
c_8	−4202893845	c_{23}	124726632332855709	c_{38}	−125214173553060603	c_{53}	−851131296
c_9	−52527801829	c_{24}	−81630702907168435	c_{39}	74148990943615773	c_{54}	−7404480
c_{10}	389012073774	c_{25}	−233304922916464116	c_{40}	1323145848555882	c_{55}	11607552
c_{11}	−739245505029	c_{26}	369208240308978903	c_{41}	−17245899318998481	c_{56}	−1069056
c_{12}	−3630388813193	c_{27}	149184938547990970	c_{42}	5998397764413635	c_{57}	32768
c_{13}	22320737197311	c_{28}	−722029723139114721	c_{43}	737303534151183		
c_{14}	−19496199822597	c_{29}	299047633459578630	c_{44}	−1004697879945234		
c_{15}	−179705526811852	c_{30}	714591055973464177	c_{45}	188962753704321		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 61$, and it is the recurrence OEIS A229636 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{57} - c_1 x^{57-1} - \dots - c_{57}$. Its constant term is $-c_{57} = -32768$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 57 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 57, so $\deg q = 57$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 57, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 57, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 32768.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-57 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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